

Jahnavi Saxena

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Education

UPES

B.Tech (Hons), Computer Science (Cybersecurity), CGPA: 8.19

2023 – Present

Dehradun, UK

Technical Skills

AI/Agentic Engineering: Claude Code, GitHub Copilot, prompting and structured tool-use patterns, eval/guardrail design

Security Engineering & Automation: Python, REST APIs, Bash, Go, PowerShell, Java, Docker, Git/CI/CD

Cloud Security: AWS (IAM, VPC, EC2, S3, CloudTrail, GuardDuty)

SIEM & Security Monitoring: Splunk, ELK Stack, EDR/XDR, Windows Event Logs, Sysmon, Linux Auth Logs, Network Traffic Analysis, Alert Triage

Governance, Compliance & Privacy: SOC 2 Type 2, ISO 27001 (Basics), NIST CSF

Digital Forensics & Incident Response: Incident Response Lifecycle, MITRE ATT&CK

Experience

Xebia Technologies

June 2026 – July 2026

AI Engineer Intern

Remote

- Built and prototyped AI-driven engineering workflows using the **Claude Code SDK and Google Antigravity**, applying **prompt engineering and structured-output design patterns** to automate development tasks.
- Used **NotebookLM** to synthesize technical documentation into mind maps and use-case write-ups for stakeholder presentations.

Infera Technologies

June 2025 – September 2025

Cybersecurity Analyst Intern

Noida

- Conducted authorized penetration testing across **10+ web applications**, identifying **30+ vulnerabilities** using **ZAP, Burp Suite, Nmap, and Nuclei**.
- Automated reconnaissance workflows with Subfinder and httpx, reducing manual triage effort by **40%**; produced **5+ VAPT reports** with CVSS scoring.
- Supported **SOC 2 audits** by validating evidence for **20+ controls**, collaborating with development teams to strengthen secure SDLC practices.

Projects

Zero Trust IAM for AI Agents | GitHub

June 2026 – Present

- Built a **Zero Trust identity system** for AI agents, addressing the fact that **97%** of AI security incidents trace back to overprivileged agent access (IBM), aligned with Cisco and Microsoft's 2026 Zero Trust for Agentic AI frameworks.
- Designed **6 core modules**: **JWT-based identity registry**, **OPA/Rego policy engine**, **Redis-backed just-in-time access broker**, and **ML-based behavioral anomaly detection**.
- Added **tamper-evident audit logging** and **human-in-the-loop approval gates**; containerized and CI/CD-tested via **Docker Compose, pytest, and GitHub Actions**.

Kestrel: AI Agent Vulnerability Scanner | GitHub

May 2026 – July 2026

- Built a **full-stack AI agent vulnerability scanner** integrating **SAST and DAST** checks, mapping findings across **OWASP Agentic AI, OWASP LLM Top 10, and MITRE ATLAS**.
- Ran an end-to-end scan on a live demo agent, surfacing **6 real findings** spanning **all 4 severity tiers**, including a **critical wildcard tool-scope permission** and a **hardcoded credential exposure**.
- Engineered a scalable scanning pipeline using **FastAPI, Celery, Redis, and PostgreSQL** to reduce API response latency by **95%** via background task offloading, integrated with a **React security dashboard** and **GitHub Actions CI/CD** for automated security testing and deployment risk gating.

SOC Home Lab | GitHub

Jan 2026 – Feb 2026

- Built a self-hosted SOC lab with an **ELK-based SIEM**, ingesting **Windows Event Logs and Sysmon** data from **3+ sources**, processing **500+ events/day** with custom dashboards.
- Deployed **Zeek and Suricata** for network traffic analysis and intrusion detection, alongside **15+ custom Sysmon rules** for brute force, privilege escalation, and lateral movement.
- Mapped **10+ attack scenarios** across **5+ MITRE ATT&CK tactics**, simulating the full attack chain end-to-end.

Research & Publication

Securing Edge-Cloud Computing in 5G

International Conference (ICACDF 2025)

- Designed a **multi-layered, zero-trust security framework** mitigating DDoS and data interception; developed an **AI/ML-based anomaly detection system** improving threat detection efficiency by **30%**.

Certifications & Achievements

First Place – HackFest 2025, Cybersecurity Hackathon

TryHackMe – Ranked in **Top 3% Globally**

Google Cloud Study Jams 2025 (GDG UPES) & Google Cloud Skills Challenge (Achievement)

Cisco: Linux Essentials

Microsoft Learn Cloud Skills Challenge (Build 2024)